ActInf Livestream #039.1 ~ "Morphogenesis as Bayesian inference...."

Livestream | *1:50:03*

All right.

Hello, welcome to ActInFLab livestream number 39.1.

It's March 2nd, 2022.

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We're here in ActInFStream number 39.1, and we are learning and discussing this paper,

Morphogenesis as Bayesian Inference, A Variational Approach to Pattern Formation and Control in Complex Biological Systems,

by Kuchling, Friston, Georgiev, and Levin from 2020.

And we had a fun dot zero and time probably all individually going through the paper.

Pretty much in 39.1, we'll just kind of go over introductions and then can go to a blank page and just see where,

especially Steven and Blue want to raise any questions and also anyone can ask in the live chat.

Okay.

So we'll just do the introduction.

So I'm Daniel.

I'm a researcher in California and I'll pass it to Blue.

I'm Blue.

I'm a researcher in New Mexico and I will pass it to Steven.

Hello.

I'm Steven Sillett.

I'm based in Toronto.

I'm sort of an action researcher and practitioner in community development and participatory theatre.

And I will pass it to Dean.

Good morning.

I'm Dean.

I'm here in Calgary.

And not much to say other than I'm kind of looking forward to hearing what Blue and Steven have to add to this,

this rather big math question and how that fits with morphology.

So back to you, Dan.

Okay.

Well, if anyone wants to start with a specific question or figure, or we can review just sort of the main points of the paper,

like let's look at the roadmap.

So what section would people like to enter in or have a question or idea related to?

Or we can just go.

Yeah, Dean.

So you wanted to talk about this and I wanted to talk about this and it was, it was this whole section on the least action principle,

which I think we both find quite interesting because yeah, why?

And it led me to the question of, is there, is there something efficient about search over somebody sort of passing along and sharing?

So I'm kind of curious what people think of that in terms of the math and the morphology.

All right.

Based action, math, morphology.

Steven?

Yeah.

I don't know if we could unpack a little bit more what Dean was saying there.

I'm, I'm interested in some of the implications of that work around looking at the math and morphology as it scales from these, what you might call smaller scales than we can normally perceive.

Often what's going on with morphogenesis is not something we can be consciously aware of a lot of the time.

However, who we are is built upon morphogenesis.

So I'm, I'm very curious around how some of these first principles can be thought of as rolling out.

So when we talk about lease action as being one way of approaching it where people often think about energy and efficiency of energy processes as being the route that everything will take,

the more informational Bayesian inference approaches and learning that is implied by this.

Without sort of distracting from the paper, I'm curious about people's thoughts about what sort of implications that has for thinking about these bottom up ways of knowing.

Okay.

Blue?

So I have some questions going back even to before lease action in the paper talking about generalized flow.

And it's really interesting.

Just in light of the paper that's coming up in number 40.

But, but here, you know, we see that this is like a spatial, like we're talking about spatial motion here, or like the spatial motion.

And I think that's the spatial motion here, or like the foundation for the way that they model morphogenesis is, you know, positions of particles through space, right?

Like, and time.

And I just, I just wonder, like, we have a time evolution here, and we're talking about motion in space.

But could we do the reverse and talk about like a space evolution in terms of time?

Like, in terms of like, can we invert space and time?

Like we make this kind of flexible, we make this

specifically at equation 15 that I've not seen before is P with a dot. And it was not defined in the paper P with a dot, but I'm assuming it's the first derivative of P. And I just thought that that was interesting. So it's the probability density, but is it the derivative of the probability density with respect to X with a tilde? Or is it just P dot because X has a tilde?

Anyway, that's my question there.

All right. Wow. Well, we have a lot kind of on the table.

I think the thing blue appended with generalized flow is a good entry point because this is sort of what the system's state evolution is. And in live stream number 26 with Bayesian mechanics, we also talked about the generalized coordinates of motion with relationship with control. So it's kind of like a thermometer cybernetic model.

If you only have the position of the temperature, there are certain strategies that can be applied given that modeling just simply the temperature. And then if you are modeling progressively higher and higher levels of how the system changes, it's kind of like a Taylor series. It equates to basically a better depth of control. And then part of the hard part is that because over the time course that action is being planned for even hypothetically changes in like the unknown consequences of one's action, complicate the estimation of different policies. So this is just not like a total perfect way to roll out the system and understand how it's going to happen, but it gives like approximation terms to higher and higher levels of the system changing. And so that's the, um, and all of the levels of analysis are being linked to each other by being derivatives of each other. So then some stationarity like is capped out at some point. So the derivative becomes zero. And then that's like a heuristically approximatable, um, depth of that model.

So that's just like one note of what is being estimated on. So it's like temperature and its higher order components in some either analytical kind of infinite generalized way, or just more practically with some smaller realization. And then the dot is like a derivative, pretty sure two blue, like it's like a prime, like prime, prime for double derivative, two dots for two derivatives.

Um, and then morphogenesis is like position of particles. Um, here it's like the cell body or the cell nuclei or the center of gravity of something. And then those are spatially moving. So that's the lowest level of this chain of the integrator chain with the generalized coordinates of motion. So that's how it's from a

Bayesian mechanics getting to morphogenesis with just how particles are moving in space. But instead of just modeling like their X and their Y, it's like X and Y and the higher level derivatives.
So that's the developmental motion of particles. Just to summarize that part before
the space and time. Okay. Any thoughts on that? Because the generalized flow is definitely where
whatever least action is, is going to be based in.
Um, okay. So then let's just look at that formalism.
Um, here is the derivatives. Okay. Anything on this? Yeah.
Yeah. Yeah. Yeah. So if you're going to look at the, at the formalism, um, did you pull it up here?
Yeah. Yeah. Generalized. There you go. Um, so go ahead and then I'll, I'll interject where
just, yeah, here's 2.2.1 X tilde is the total blur of all the dots on top. So yeah, go for it.
So just here, the, um, my, my question more is the noise and fluctuation term. Um, and is that just
like Brownian motion or I don't know? So yeah. Like what is the, where's the Omega random fluctuation?
Like, is it just, I don't know, random in, in terms of Brownian motion, or is there some other
random fluctuation positionally?
Like, do things wiggle?
Definitely someone who knows more about, uh,
uh, stochastic processes. And I know several people listen who do, um, could help with this,
like in a math stream on a little bit of the stochastic aspect here. But I think at a first pass,
um, there's situations where the, what is modeled as noise versus what is modeled as signal have
differential ratios. So there's times where like the regression is very, the F, even if it's a
regression and other functions like very tight. And then the noise is modeled as low versus
one where just the location of X is just dominated by the noise term, but I'm sure it's way more
than that too. Um, Steven or Dean?
Well, I was just gonna, you know, often when we see, I'm just thinking broadly in terms of
the modeling approach is an active inference. Often it brings in temperature as a way to sort of bring
in the kind of noise fluctuations into models. Whereas here it's directly seen as a noise fluctuation term.
Um, so I think that's, uh, maybe it's again, because we're going down to even more first principles here,
um, uh, it's getting down to stochastic dynamics, which isn't necessarily even thought of as temperature,
but as something, um, more, more foundational.
Okay. Um, yeah, Blue.
So temperature, um, is like, yeah, right. Like with increasing temperature, um, like
random motion would increase. So that does, um, make sense to me. So another, um, point out of this
section, or maybe if anybody else has a point here, um, going down into the next like set of equations,
unless anybody else has something else. Yeah. Just to summarize that it's kind of like the
temperature parameter is understood sometimes to play this role of mediating the difference between
the particles motion being driven by some sort of directed motion, um, like change in position through
time, not at stationarity of position, which is like kind of the observable in that chain that directed
motion versus thermal slash Brownian. And then, um, that might be kind of where these statistical

assumptions come into play. So it's like variance estimation, but temperature is just one kind of
variance with the move, the motion of particles, but, and it has a, it has a physical interpretation.
And I mean, it's not an insignificant piece. I don't know. We could explore more where temperature
specifically comes into play, but Steven, I think you're getting at the right thing that this is not
temperature bound, but it does relate to that sort of situation. Okay. Blue. Um, so I don't know if you
want to go down to equations, um, 15 and 16, probably where my next question or comment is.
Do you have them in there or no? Oh, copy it in right now. Cool. Go for it. So it says, um,
equation 15, uh, I think that's X tilde dot. Yes. P X tilde. Um, it is a probability current,
which I thought that that was cool. Cause I've never really heard of that like term of probability
current, like how probability changes over time or, or, um, how the probability density changes over
time. Like what is that? A probability current. Like that was just, it's an interesting, I don't know.
Um, like term that I hadn't heard before. And then in the second, um, equation 16,
it says that, um, this is a partial differential equation that describes the time evolution of the
probability density under dissipative and conservative forces. So where the term on the left is the
dissipative forces and the term on the right is the conservative forces. And I thought like that,
that was interesting, especially relating back to the dot zero video, um, where you guys were talking
about a tensor and Dean was talking about tension. And I think that there's, it's like an interesting
tension here between like conserving and dissipating. And it just makes me like wonder how, um, how like,
like is death when like the dissipative like term is winning? Like, is that like what causes like a
system or, I mean, I know that the, the term, you know, winning or term being greater doesn't cause a
system to die or when the system is decomposing, if that can be represented here in terms of these
dissipative and conservative forces, um, um, in the process of an organism dying or ceasing to exist.
Okay. Sorry. I have way more questions than.
No, nice. Nice. Let's, let's look at 15 and 16 a little bit. So yes, probability current it's kind of
in this dynamical systems framing it's being bridged with formalisms that are like more like a current
like a flow on a dynamical landscape. There's that sort of angle, but then there, um,
and maybe there's also a more of a fluid flow component blue.
It's a time.
Sorry. I was muted. So like, it's funny when I look up a probability current,
like also foreshadowing, foreshadowing the dot zero, it says in quantum mechanics,
the probability current sometimes called probability flux is a mathematical quantity
describing the flow of probability. Um, so, so it's interesting that it's related to quantum mechanics
and in terms of these, like, instead of these classical physics, um, things that we've kind of been
looking at or dealing with so far.
Hmm.
Yeah. So this 14 is
Okay. This might be a, only a partial
element or, or example, hopefully one that is useful. So this is X tilde with a dot. So that's X tilde,

all the coordinates of motion, and then X with a dot means derivative of, so this left parameter
with a tilde and a dot is how the generalized coordinates are changing through time. So that's
kind of what is being modeled. The data that are being, whether they're an observable or an unobserved
estimated state in a Bayesian graph. So whether it's an O or an S, that plus it's all of its derivatives.
And then that is being partitioned into one big complex function, F, bottling it, flow.
And then this noise, omega tilde, over, both are over tilde. So the, these are both over,
like each of these higher derivatives also have an associated noise term.
And then it says, yeah. Okay. Steven.
In some ways there could be an analogy. I'm not saying it's the same, but there's a,
there seems to be a mirroring of the enthalpic turn and entropic turn that you get in Gibbs free energy,
where one is where there's this, the, the energy is bound within the molecules at play.
And then the other term, this entropic turn where it's kind of dissipating. So, um,
there, there, it's, it's another, because these things are, are, is translatable, just like mass
and energy are translatable and they're known to ultimately be conserved. Um, they can only be,
the overall mass energy of a system is always conserved, but it could be translated between.
Normally we don't think about that because it only happens in nuclear reactions. But if you're thinking
informationally, it's still, and this, and I see they talk about this idea of probability mass.
So that there's almost, uh, an ability to think about that conservation of, um, energy in a form,
which has mass, which has form as opposed to when things are dissipated as light, where it doesn't have mass anymore.
So I think where they're talking about the conservation of probability mass, I think that they mean like
the conservation of the total probability there, like, so the probability mass function, right? Right.
Like, like, so I think it's that, um, which, which relates to like the, it's the conservation of mass,
but it's also the conservation of information using that conservation of probability. Like the probability
of all of the possible things happening has to add up to a hundred percent. Like that probability is never
going to change. Um, and so I think that that is there.
That it has to add up to one like a fraction or a hundred percent, uh, has the realist and,
and instrumentalists take the realist take is something has to happen. And then the instrumentalist
take is if we model it this way, we're going to be able to use statistics. But if it was like,
well, the chance of getting heads is 60% and the chance of getting tails is 60% in this next flip,
it would open up the space in a different way. Um, it it's, it's kind of outside the bounds of the
model. You could have a model where both happen, but still there's, it's a different thing. You know,
can you model things that can't happen in the real world? Or if it's not a good measuring stick
for statistics, which is kind of like that one, something must happen. We have to model something
happening. We have to model a dataset row being added. Otherwise what is happening if it were like
one, one, every one, every 1.2 rows. So, um, and then here, yeah, Steven.
In relation to the point you just made, I'm just curious whether if these functions can be used in
multiple particles. Um, in say for instance, you have the pro the idea is the probability of a coin

being flipped is 50 50 for instance, or a rigged coin could be 60 40 in one way. However,
the probability if there's noise on how rig something is, so for instance, you've got,
you've got a coin and it's flipping between being rigged 60 40 50 50 40 30. So there's kind of a,
there's a random fluctuation in that behavior, which may be contextually contingent. That in itself
wouldn't be in the one equation, but if you have multiple particles there, they could start to behave
not just as one predictable thing. They, they themselves could have the change in probability approaches.
Thanks blue. So I think like specifically as related to here, and I think that this is derived,
like I'm not, I don't like really know. Um, I don't really do these like derivations. It's definitely
not my area of expertise, but I think in, in terms of the conservation of mass, like, like we're not
creating or destroying matter. It, the conservation of the probability mass, it just says that all the
particles, if you're looking at, it doesn't matter if it's one particle or a thousand particles,
the particles are somewhere, right? Like, so the particles are not nowhere, like the particles are
somewhere. So, so it's that kind of concept, or that's like the idea that, you know, all the
probability has to add up to one, like there is 100% chance that you will find the particle of interest
interest somewhere. I mean, you're right.
Okay. Um, yeah, blue, good section to, uh, pull out important, some important bridge equations.
So we have the signal to noise. Now, when the signal that we're modeling with the instrumentalists take,
we're modeling how much the cow weighs, defined as the one that comes to walk over to me when I call
for it or something like that. That could be dominated by different functions or could be
driven by different noise on different time scales. So the signal sort of directed motion component or
modelable component of the motion through time is the purple. And here is like some
flow with the upside down triangle that we talked a little bit about in dot zero, but it would of course
be good to hear from someone who knows more about this flow, um, operator object. Um, signal is like the
movement on a probability distribution on those states changing. And then the noise is the dissipative component.
Steven?
Steven, it appears as well that by breaking it into dissipative and conservative components, it's, um,
it's serving a similar role to separating into complexity and accuracy. It's even a way to, um,
break apart for dissipation, um, where the, um, because you're not
actually talking about a final target, but more like you things just doing what they do in some ways
and having a very, um, kind of physics fundamental process. I think that's proving to be useful because
it's also, again, something which can be translated into different, different types of, um, math, different
types of physics.
Okay. So let's connect it to the morphogenesis. So what do they do in the paper? They're modeling
the motion of cells. So why does it matter this whole discussion that we're having about how to get
from the motion of generalized coordinates? If you're the modeler, you get to define that and get to set the
parameter between this. If you're the experimentalist, then you're inferring this kind of like a regression.
Like if this was a regression term and a noise term on the regression that, and then there's some

function that helps you fit a certain solution to the one linear aggression that fits it best.
So it's like kind of in that same genre here. So if we take the realist take, um, on that, it's like how the actual system is defined as, or inferred to be as from experimental data, and then flowing the probability density on that of our inference on that, whether we're getting that from the data or whether we're defining that system formally. So that modeling, we could help from the authors or someone else, like
it engages some new future. It's like a bridge to some other set of equations. So that's where it would be good to learn more. Like what is actually enabled here, but it enables a new Vista that separates that, that enables this separation. So we can talk about, instead of this, like potentially extremely open-ended and non-linear change on the system can be bounded within a statistical inference, zero to one inference framework that is more amenable to certain kinds of analysis. So that would help model cell location and all of that. But I think there's still a lot more to learn there and unpack. Okay, Dean, yes.
Dean, I want to just sort of step back for a second here, because I want to go, we started out talking about, so what are the implications for least action? So the act part, and then we sort of stepped
into the idea of what, well, what came before the action? Well, it was these flow states. Now I kind of want to go back to the action part in this, in the, under that section 222, the second paragraph says that the least action principle can predict the emergence of form. So what Daniel was mentioning there, that sort of the morphogenesis. In terms of the flow or paths of least action in biological systems,
for example, in colonies, ants find the paths of least action to harvest food and bring it to the colony.
The example considers their paths as flow channels or trajectories, finding the least average
action for each instance of foraging given available resources. So this gets me thinking,
about what do we mean by least action, as opposed to say, most action, and that general way, the general way that things tend to follow. Like if you thought of an analogy of maybe a 19th century war, where armies would line up facing one another, parallel to one another, this is speaking more like a flow state, more like you find in nature, where the molecules we talked about here are sort of flowing and following down the side of a bowl to the least potential energy. And so now my question is, okay, so why does nature nature tend to flow with the advantage of least action, whereas people with instruments don't necessarily flow or follow one another unless they're mirroring some other form or some other shape. So
that's interesting to me because I think by nature, we are quite adaptable. But when we start making non-living things as instruments to necessarily take the nature away, the life away, we're maybe not as quite as adaptable. So I'm trying to bring it, I'm trying to step back and now pull us back into this idea of, so what advantage is there in this physics view of least action? There must be something advantageous there in nature. So what is it? I'm not the geneticist, so I don't know what this signal milieu is
that makes that so. So I want to tap your guys' and gals' expertise on this because I don't have it.
All right. Thanks, Dean. Stephen?
So following on what Dean's saying there, I think in the traditional sense, we tend to take what we
call equilibrium dynamic approaches. So for instance, the armies are lined up, like Dean said, then there's a battle and then there's the final state. So we talk about the initial conditions, which is equilibrium, everyone lined up, and the final state. And then the bit in the middle, we sort of talk into our

shirt or our shoulder or whatever, cough a few times, and somehow it happened. And in some ways, with the biological approach, it's like it's in that, because all that bit in the middle is where the flow is happening. So essentially speaking, and biology doesn't have those initial conditions in the sort of traditional sense of pure equilibrium, it's always in this non-equilibrium, or at least a large component is in a non-equilibrium state. So this is an interesting thing is it's flowing down towards the least energy. It never reaches it. So it reminds, you know, I think that's, it always, so for instance, in the chemical reactions, you set, you take your pure reactants, you mix them together, you stir it up, you do the reaction, and then once you've got your product, which could be a precipitate, for instance, which, you know, it could be a ton of the precipitate, it then gets filtered, it gets washed, it gets dried, and it gets measured for how pure it is. And now it's in this kind of stable product form, you know.
So I think, yes, this question about what does it mean to flow is really important.
Thanks, Stephen. All right, Dean?
Well, what this flowing and following thing kind of says is that the desire line or the termite mound, the result is of signaling, which is, I don't know if that's collective signaling, I don't know if it's the stigmarchy part of it, like I don't know why nature sees the advantage in the following.
But there has to be something there, why is that advantageous? Why is that more adaptable than the alternative?
Okay, thanks, Stephen.
Well, I suppose one way of looking at that is it's the only,
it was, I mean, if we're going to take a realist route, recognizing that it may be,
it's the only plausible route on the table.
So we, they don't get to, you know, forge teeth in a kiln or, you know, in a way,
that's partly why all chemicals that are available in nature have to be within certain plausible temperature scales and pressure scales that can yield those products at.
So, I mean, that kind of doesn't answer the question exactly, because I know that's kind of a bit of a, but it's part of it's what's available, I think, in terms of biological plausibility.
And I suppose, in some ways, humans, we've adapted a cognition to try and move outside that.
Okay, yeah, Dean.
Now, one last thing on this, and that is that following implies that there's also something leading, there always has to be something taking that, that initiative for others to fall in line behind. So, shifting that and making leadership something that's more lateral, so sort of going forward together, as opposed to, who do we, who do we decide to, to follow behind, that's an interesting thing in terms of sort of sussing out how this least action principles thing turns into things that we actually see phenomenologically. So, again, we're going to get into the, what the final form takes. I just think that when we were sort of treating this in the point zero, this was kind of a pivotal moment, because it went from, what are all the things that have to be in place for formation to occur, to now, let's look at what the, what the form now results as. And as I said in here, in that one paragraph, it said,

well, the previous paragraph says, physics offers a useful formalism to understand at a quantitative level, the ability of biological systems to work towards. So now we're, we're basically now moving to something that is, that is adaptable. So I think, like I said, I don't want to over,
or talk this one point, but I think it's an interesting part in terms of, especially when we
get into the, the discussion at the end, around why, what, what some of the assumptions were and yada yada.
Okay. Thanks. Yeah. Stephen. I suppose this work towards, um, idea. I mean, that can be
thought of in a, as a modeling and a life realist problem. Like how, how does that work towards, uh,
and in some ways, uh, accuracy and complexity can
be used once we, if we're at the level of assuming that we are actively inferring to stay alive. But
prior to that, something like conservation and dissipation or conserved are a bit more foundational.
Okay. Because if I conserving or working within this conserving, dissipating noise dynamic,
at the levels, which are below what we are consciously aware of, however, which all our
cognition is effectively built upon somewhere, we, we, we're able to potentially
give that towards this without necessarily having set a goal. And it could be, and I think that's,
um, that might be an interesting, see how that rubs out in the real, real world of, of, of active
influence, labs and papers and stuff. Um, yeah. Cool. Yeah. A lot. You linked there, Stephen, just now, like, accuracy, complexity, and then, okay, we'll leave it for a future day and work. And who knows how many of these are just concordances versus tautological, but accuracy within a model is like pragmatic aims within a model with the imperative to prefer to fit as much data as possible. And then it's like, um, conservative, um, using the reward structure that worked at one time step, using that generative model moving forward versus changing it, any sort of dispersion around the parameters as evidenced at the lowest level of the chain, the position of the particles or the actual parameter that's being modeled, like the actual image that's being percepted on versus these higher generative models, which don't realize like at the kind of, um, you know, tip of a javelin, so to speak. So then the signal and noise is also like conservation and dissipation with respect to the model. And then that is coming all the way back to morphology with like the position in stasis versus vibration of the particle from thermal. So
so it's like a lot that gets linked here, but where's least action in all of this?
I, yeah, blue.
So just to read a quote, um, from the paper, it says, um, since self-organizing open systems are
not conservative, their structured flow is quintessentially dissipated, dissipative. And so,
and, and that's like, you know, down it's in the third paragraph in the least action principle section.
So it just, that goes back to what I was saying, um, earlier, like, does that equate to death? Like
when the system totally dissipates, like it's ultimately dissipative. So, you know, we stay in this,
like nest for a little while and then we dissipate, right? Like, is that, can it, can it be modeled?
Like these processes using these equations? I find that super interesting.
Yeah. Um, thanks, Steven.
This, this, um, this idea of the least action, I think tying in with what blue was saying there as

well is, um, um, it's least action. Maybe we should say dynamical least action as opposed to least action, which we normally think of IE. Well, like I was saying, reactants products, what was, what's the, what's, you know, what's the route to get from A to B or the army lined up to final space.
Um, normally we are taking starting conditions, ending conditions, um, or, you know, where I am now, what's my goal, what's the least to get there. But those two things are defined in some ways.
They're, they're, they're definable. So I think it's a good, now, all the bit in between is, um, um, trying to go to least action, um, in some way, um, using that kind of in, in the space of chaos,
I suppose, if we're going to take the kind of complexity approach in, in human systems, um, it's in a chaotic state where you're trying to at some time, um, make sense of how to act. Um, but however, um, you know, in some ways there's only chaos moving into complexity, which is available at some biological systems, theoretically, maybe not the idea of complicated and kind of simple sort of reproductive sort of steady, like equilibrium state systems, like mechanical systems aren't necessarily available. Although you could argue that maybe certain properties of our morphology, like our, um, our, our bone structure in that sort of gives something which is approximate to that.
Um, once we know how to move our arm, we have a relatively simple thing that we can now bring into our regime of attention. But, uh, yeah, I wonder what people think of dynamical least action is, is somehow what's going on here.
Yeah. Thanks Steven. Just here's one take on that. It's a good idea or question. I hope this is accurate too, because I think we're all learning here, but this is definitely a really challenging area in some ways to approach, especially given our realistically limited familiarity.
So again, it'd be awesome for people who have more familiarity with these equations to join us like either in preparation or even just joining us on these streams so that we can actually learn and connect to these other areas. Somebody who sees this formalism like every day, just like we might see some other one. Okay. But Steven brought up, is this a dynamical least action? Okay. So first kind of a hopefully non-contentious point that we're analyzing what we model. We can only ask the computer to calculate the numbers that we ask it to model and add. We can't expect something that goes truly beyond that. Not to say that the models can't have surprising outputs or interactions, etc., but we can't have it go beyond what we specify. And the model is over x tilde, which is the generalized coordinates of motion, which is the position and all of the higher derivatives, the whole integrator chain.
Position and all higher order moments of the statistical distribution. They're called moments, like the first, second, third moment of the derivatives of the statistical distribution.
And these are basically terms of approximation of motion that allow a high order, like a Taylor series approximation or a Volterra series approximation to allow snapshot modeling with real time flow.
So action, cognition, perception. It's the flow over, as gets explored, the blanket states. This is like the flow over S,
A and I. States, actions, and internal states. So like this is enabling a flow description of Markov blanket states
and their perception, cognition, action, their flow over states, their chains through time,

including higher order moments of time, so that it is a dynamical systems model, but still there has to be a snapshot. Like a time series model of a stock price, it has a value at whatever time resolution or continuous or discrete, it's being time series modeled at. So it's kind of the relationship between snapshot modeling and capturing higher order trends. And this is a certain way to think about that in the direction that they're going to take it towards the Markov blanket partitioning and bridging it to everything that that affords. Okay. Dean and then Steven. And then both.

So this is really important because I think it, I want to tease something that maybe we can look at right, right here, but only in the point two. So between the us doing the live stream zero and today, there was a report about the rate at which climate, I think it was a UN report, the rate at which climate change is happening so rapidly that as, as the people who are sort of in that highly changing flow state aren't able to necessarily going to be able to adapt to the rate of change. And so I wonder if some of what this paper speaks to might help us understand why the authors of that report are indicating to us that the, the environment in which we, we are existing in is going to change so quickly that as cells within that larger structure, our form is not going to be able to adapt quickly enough to the way that the external system is, is changing. So I want to save that for the point too, but I think this least action principle part would be a good entry point into some of these bigger questions going forward. So I just want to park that, but I think it matters. Yeah. Just to give one note on that, the rate of change and how it changes, those are kind of natural language descriptions of derivatives of how things are changing. Might be a simple claim to some, but that's really important to keep in mind. So how things are changing and then that's always going to be unknown to some degree. So how things change and so on, this is the generalized coordinates of motion. And so it is about modeling rates of change and predictability of systems that have rates of change. And okay. Yes. Steven.

Steven

Steven

Steven

Steven

policy model okay so we've got this recognition coming in but it is derived
by this action and of course one challenge you've got in the climate
change scenario is really what you think is one thing but being able to act and
being I mean the danger is we hit a point where no matter how much we know
it's beyond the ability to act in our capabilities and
capacities as a species the so this this way while I think the word cognition is
useful it may be useful to be careful just that the inference process is more
general because it can be it can be a process which is like say happening
beyond thought so thought is a is another action almost given a higher order
understanding on the generative motion project generative model for an action
in active inference of course a lot of what's going on underneath is beyond our
perception literally way beyond so that that that can be interesting or useful
to strip out nice yeah blue thanks so just to kind of go back to dynamical least action I do think that
that's what they're referring to at the very end of the section the authors say from our perspective
the key observation here is that any dissipative random dynamical system can be formulated as gradient
flow on the log likelihood of its states this is reflected in our solution to the Fokker Planck
equation in 17 which means the action is the time or path integral of the marginal likelihood or
self-information for any system or model M so this is really the key thing this means the least action
integral over the Lagrangian turns into an integration over the self-information of states which is known as
entropy in information theory in short the principle of least action manifests as a principle of least
entropy for systems that possess a random dynamical attractor and thereby obtain non-equilibrium steady state
thanks a lot blue great point good day France hi hi sorry that I'm late and uh
uh I only have a half an hour or two I'm in the middle of finding preschool so daycare is a whole
mess right now I do apologize I don't want to take this time it's a little bit of a mess next week should
be better I think we found a school so hopefully next year we're going more easy cool unexpected but
you know how will we model that but thanks a lot for joining this is really cool so we were just
describing some of the formalisms but where would you like to begin it'd be awesome to hear like any
just introduction and context on the paper and then we kind of jumped in at least action and would
love to hear your take on that sure um all right so little introduction so I guess um I'll just start

all of myself so um I work in my 11's lab and Tufts and doing a PhD in biology and this is the whole kind of active inference um part of my work is came actually before my entire PhD so I joined Mike's lab so I wanted to do really learn what kind of information and physics do in a biology context so I didn't I didn't want to study the whole level of you know I mean I'm still doing it as a technique but I didn't want to have the focus on protein interaction and you know genetic information um where you measure all these things so I was basically wanted something a bit more broad than that and so when I reach out to Mike and once I got some accepted he um I was like you know have some months to kill can you send me somewhere or do something cool and then he sent me to Carl Frist and that's when that's the first time I heard the term active inference so I spent about four months in Carl's lab um and then basically started working on on on the I think half of half the work in the paper I was actually done in those those for a couple months and when I was in Carl's lab so the the overall goal of the paper and of that part of my work is to see if we look at morphogenesis and similar biological processes can we look at something where you have a very kind of um kind of baseline stem cell like behavior where you have cells that can't do like morphogenesis basically I don't know how much you're familiar with the whole biology aspect of this paper but essentially it's the idea is you model cells that are uninformed they have some kind of you know genetic code something that encodes their their their structure the the basically you have the generative process already ingrained in the generative model into the into into those cells but they're completely where their stem cell like they don't have actually yet achieved their final form um and they are just like in the simulation in biology right they're starting off with very few asymmetries this is I think there's a famous I forgot who said that but uh some physicians at once said most of physics is basically just uh symmetries it's symmetry everywhere and that's exactly kind of um how most biologists think biologists think of of morphogenesis there's some asymmetry in in the beginning in the egg or some there's some some different information some gradient of something some localization of certain agents and then from that on basically everything else just follows through but it's it's it's hard to believe that's the whole side of the story because there's just so much complexity and so many cells doing this at the same time so if there isn't any capability for them to adapt to signals in the environment and to learn from each other it's hard to believe that the time frame they're given they can really um you know achieve the complex morphogenesis outcomes that we know um do exist so coming to this

paper basically this model is actually already um like the the father or mother of this paper was already done before i even heard of this topic that that was the annoying one's place so the model structure itself was already done by i think called it most if not all of it and mike just basically was on the paper and giovanni as well they basically gave him some input on on on what what the background of biology would look like and um so they already figured out basically let's do a simplistic model eight cells there's more as far as i know from carl and as far as myself in my simulations many practical concerns don't want to complex if you have too many cells at once um then then you're going to problems in the beginning something i i actually add on to the model um even the baseline model was that damping parameter um which i talk later on in the results which might often

like it's not it's not a key part but i find it very interesting because it's one of the requirements for
for active infants um for for free energization variational management organizations there's a
smooth landscape and when you have all these cells initially clustered together and they're trying to
infer their place and they all kind of have some random randomly initialized prior beliefs they don't
know where to go from there and if you right if you have already high precision in there and their
sensory apparatus initially they have a really hard time for they still end up mostly fine getting
later but they jump all over the place and you can imagine that'd be very bad for an biological
scenario if every cell immediately jumps to the first queue they have we actually think now in the
mike levin labs that um love cancer initiation happens a little bit like that where positions are
as i said too high um there's a paper coming out soon by um by a colleague of mine that talks a bit
more about that argument but so basically from um we had the target morphology already um colors encoded that and
you saw what you see in the in the baseline control experiments was already done what i was interested
in is basically again coming back to my my question and my approach to this from the biology side of
view is how can we use the aspect of information flow and specifically active inference to manipulate and
and better control morphogenesis outcomes and control biology so the the two main results in that in that paper
are looking at if you set up you know instead of like a normal biology experiment where you basically
interrupt one part of the machinery um and then see how everything reacts you actually control the
process information processing itself so one was basically if you put in a an asymmetry and the response
of the cells to their to the signaling um ligands they get the signaling concentrations in the environment
how can you basically completely remodel the entire morphogenesis outcome even though you actually left
the entire coding itself that's always one thing i tried to stress in that paper the target morphology
and all the simulations is exactly the same none of them they basically said there's that coding
um of what where they're supposed to be but how the process information has been changed
um so the first figure is that where you have those two heads and two tails um and if i did the two tails in
that or if i just did it myself and didn't publish it i think it's in there as well um then that's
basically something that we also see in the lab i think that was mentioned as well that that's something
that that mike levin lab has done with an area where they basically induce those two head types right
that's something that's really weird to a biologist to and again in in the type of manipulations that we do
in a lot that have been in this lab they didn't manipulate the genetic code again the the DNA from those
spinaria that was exactly the same but somehow they perturbed the actual bioelectric network
um itself so basically on the state space essentially so that kind of like where this inspiration came
from and then the second part where we the the malformation or i also talk about cancer information
and of course it's it's can't completely called cancer because we didn't put proliferation in this in
this simulation so that's it's the initiation stage but it's the idea is that again and there's something
that we talk about also in the lab and with carl and many emails afterwards and it kind of also goes
back to this whole idea with um how how different psychiatry disorders work in the brain where you have

some of these inference processes being disrupted and then lead to large-scale outcomes but they kind of start somewhere so idea was here is that if we disrupt only the um the information flow from one cell to the other cells and the other cells to that one cell then what will that what would that cell do so basically just completely reduce its um its sensitivity to the environment um and then saw it happen and what happened was that this one cell basically kind of like it didn't it didn't move a whole lot it got inferred the wrong kind of cell type um and it was completely out of place right so that's always bad in biology if a cell is doing something that's not supposed to in a place where it's not supposed to be um and what i thought was interesting afterwards well okay so how can we rescue that and again like right not trying to rescue it in a traditional cancer therapy idea where you bombard the cell or just kill it how can you actually have the system remodel itself so that the idea was to increase the um so the sensitivity was still disrupted from that one cell but then the flow like how much concentrate like how much the other cells reacted to it so basically we manipulated how the information flow from the other cells to that cells was flowing and increased that so and then what was interesting is that the kind of they were just kind of like going more close to interaction which is seen this time lapse figure um and then eventually they reshape so even though the cells like normally if you run this it's a deterministic simulation right so like um it's ran it's always the same random seat so you run the simulation and you run again there will be some outcome so you can kind of know which cells go where even though you know no cell starts off with like knowing i'm supposed to be in that cell that's not how it happens but because we run the simulation with a fixed randomized seed we always know which cell ends up going where and then in that cancer simulation that was not the same anymore so actually even though at the end after that rescue experiment they ended up with a perfectly normal shape in the time followed in the mythology at the end it wasn't what normally was supposed to happen right the cells that normally have gone to their fixed positions didn't all do that so there was some reconfiguration which is something that that's that's an ideal strategy that you would like right something that you exploit the systems interaction itself to then rescue it and overall target more you want to work on the the phenotype but you want to work on what's actually wrong you don't care exactly what cell does and and is implemented there's a lot of cool science low tangential but that's really cool work by eve martin that works on um neuronal networks and lobsters where they see that this works in the last course it's a simple enough system to work you only but what they're seeing is that there's actually a multitude of implement implementations that the neural neural neural network can can use and do use individuals to use entirely different parameters with like two to four four differences in ion channel concentrations to achieve the same end result and it's actually the the response to resilience to stress to perturbations in the environment is then in the details but in the end the actual homesthetic aspect of it is kept the same right the goal of having something um end up the same steven would like to speak uh go ahead oh hi thanks yeah i was just gonna one question is the idea of flow in a way more flow is better even though that um so unlike i say a generative model where you maybe don't want too much complexity you can have a lot of flow but as long as that flow is not noise um it just is able to inform so that's the first part and the second part is are you saying it's with this flow it's about finding where the choice point is or where the threshold is to say okay that's now going to decide that that's going to create a morphological target okay as opposed to it just being a gradual gradient descent on energy it is more the gradient aspect so the it is i would say it's it's not ideal to think about decision points even though in the end it doesn't happen it's something that we're just we're used to in our level of speech but on the level of of outside the underlying cells it's very much a gradual process right there's basically you have all these different probabilities which are probabilities of being one cell near the simulation and the idea of increased information flow or increased sensitivity whatever aspect you're modeling and manipulating or precision as well is that you are trying to get the vibration of energy to be minimized but also specifically to converge on a system where those probabilities actually end up somewhere meaningful right where if you don't want to be in a state where they're constantly fluctuating constantly randomized so either you mentioned noise that's one aspect you want to avoid you also want to avoid that they basically um do this and like that that that's where you're right the decision point right like the cell is basically going towards one state and then there was wrong one and then you increase the information flow of the other cells from the cancer rescue from simulation to bring it away from that but you're not you first of all not actually working on like you know identifying that point and then you're fixing that you're basically just realizing that the the simulation overall basically you basically look at the the probability changes over time so how are they updating and how fast are they updating and then of course in the end what are they updating towards and that gives you a cue if you think from an experimental point of view or even like experimental assimilation point of view then that tells okay so um if if that happened too fast if there was mis- malformation happening too fast then you probably want to act and you want to increase sensitivity or something um early on but you don't do it at one time we just set the parameters for the whole simulations but it tells you basically the strength that the sensitivity is with respect to how early and how strongly that misinformation is happening does that answer your question somewhat yeah i think that does one i suppose one question just on that is is it is there like an optimum level in terms of this flow um i it peaks or is it like the more flow you can get the better it's just literally a limit on how much information flow is possible um i think i'll have a second um so the short answer is that that's the boring answer is that always will depend on the context i would say um there is no i i don't think you can say you know more like a broad statement like more information flows always better doesn't make sense i think you already gave the answer to that why that is right because you in the end it will end up too much noise based in the system so that answer has to be respect of how much information actually is being can be processed what is the time scale of the sensor apparatus um my experimental work works a lot on um um basically having different uh variation of input signals to my model system which is actually an algae i won't get to that but long story short i actually had once got the question that where i was had these randomized signals um that was feeding it um and then someone asked like well you actually this is actually taking more information because you're having and i was like well and that like white noise technically carries more right more variation but it's not informative right so that's more information flow by itself is not better more informative information flow the informative part being is actually does it how quick does it change over time so there's a lot of the important aspect other derivatives there and with respect to the also the sensory precision and you know the timings there's certain time scales involved in like how sensors how fast sensory states lead to updates of internal states all these things together make up basically how fast an agent can react and uh how much information it can process and that's the the optimal level will be dependent on that essentially okay awesome dean or blue before we have anyone go go to the right uh so i have a question um can you flip to figure four daniel please yep um just so you had mentioned earlier france hi by the way it's nice to see you i'm really glad you could we i'm glad you could make it i was it's a pleasant surprise um but here in figure four you mentioned that you didn't there's a lot of different model proliferation in this uh model but it it seems to me that there's like more like it looks proliferative here so is it just like a very dense cluster at the first time step or do you duplicate it every time step or like how how did you end up with so many more um like beads or or whatever you know agents uh felt at the end then at the beginning there might be an optical um um so or communication problem my part so the the the the actual cells are eight in each of those images all of them but what you're seeing is the trace of it right so only the ones that have like the the the strongly colored dot and then like the little star around it at the end right if you look at the last frames on there those are actually the the cells and what you see in the back is basically like time lapse kind of like snip snip shots they're not actually cells in there uh it will be actually interesting i know you thought of that that would actually be a fairly simple uh introduction at each of those point steps you just make that a new cell that would be pretty cool actually um i i know i think basically carl and one of his when he initially published that i think he had way at the end of the simulations he basically added some new cells but he never went further with that and i think it's the same problem that i mentioned that when you have it initially and like in the first time frame at t equals one that if you have some excel close together it's um you have to kind of set the sense precision lower because otherwise they're too close together i mean it works somewhat but like it's it's very susceptible to to perturbation in that stage so in order if you basically want to reintroduce perfluoration either like you said or like like you gave me the idea to do it in between or you do at the end you would have to come all that which of course what cells do right if you have new cells formation that are not going to be the same kind of uh sense of rest that then a mature or reform cell is um but that was basically i think just for the simulation what was making a little bit complicated but um it is i'm not right now actively working on that simulation anymore if i did um that would be great yeah maybe i'll come back to that no that would be way cool and if it needs to be like less sensitive at the beginning like you just double the or yeah you just double the sensitivity at each um time step and then that way like you know you would get that enhanced sensitivity over time and it's also very interesting from the point of view fractals and there's some really cool right right now also done by i recently read that wolfram the guy behind well from alpha he's like signing this whole new physics approach where they basically just have a simple set of rules and then they just use fractal kind of multiplications and on each each time that that goes on and they create all these kind of physics laws um which i haven't yet gotten deep into i can't really say anything about how good and useful that is but of course factors we all know is it's a very kind of informative mechanism that's used in biology extensively um and that's something you could use here as well where basically each step where you introduce a new cell you kind of just have the same rule set with the initial and you run like a small simulation in that simulation essentially and with the same with a simple set of rules you could probably make up a whole much more complex so it probably looked nothing like this i wouldn't expect this to then afterwards still have like the same shape but cool stuff cool idea it's about not to have time plus right now but one day well just kind of unpack that because it's a great suggestion and thanks for kind of giving your take on it too friends it's really cool to hear like when the cells divide you may need a first symmetry break or to introduce a symmetry break you get some gradient it could be gravity it could be a nutrient gradient it can be the entry point of the sperm that triggers like a calcium wave but it's like asymmetries can give rise to asymmetries and then interacting you get two morphogens and then now it's high high and low low and then the alternations influence gene regulation and that's like very complex but um adjacency effects are really important and that's how we get the self-organization like of the insect eye or of tissues because they don't have to do what this challenge is which is sort of like getting formation spaced out this is like the bird flock morphology which is also so cool because it does apply to other systems too which i'm sure we'll be exploring more but like bodies fill out into a morphology not just dissolve this is yeah cool though i mean of course we really learned a lot let's return to least action just and feel free to feel free just to leave any time you'd like but on least action where does it come into play or what is it doing here yes so it's it's there was more of a background um section so um georgie gogiev was also on that on that paper was also my um um excellent pretty supervisor um we kind of talked to him because and this is not nothing that i did and there was like novel um i didn't like it was not with the way we framed it but i didn't make it i didn't make new math for that but the idea was that is that what was georgie um who's doing all these action principles you know the classical action principles apply to about the questions and he was like well you know it's really um i think you're missing like a communication um aspect of this where you're not going to communicate clearly how this fits into these action principles um and you're not coming clearly of of what like where diverge essentially and um carl actually did write a paper on to where the he sprinkles in there but i wanted to make this more explicit essentially that that the variation of your aviation principle is at least action principles by definition essentially but what i was trying to show here is um is simply kind of like the whole idea of this paper from from background was that you could anyone could start from this that has more of a physics background and then can work themselves to through active inference to biology and it's that's something that's physics are very familiar with right and and the idea is actually where does this um what is the interesting take on it from the from the from the variational free energy and that's what um i think in the so you start with the general definition what is an extrema and then you kind of see later on that the the whole definition of the um of the club but clever divergent that's innovation of energy very much corresponds to that and it's also um just interesting to see

basically where where it takes from so that that's now that answers your question that's basically
the the point of that was more not not that least action that i used that particular equation i did
use the classical uh the typical equations that carl uses for his variation of energy there's more of
motivation and i hope it goes to show us essentially where this all flows from like how how do you get
from from you know from from from from the up-run function in the beginning possibility how or
the branchian function how do you get from there to a variational peer energy to really make that
um kind of integration to physics tighter which i know i failed at because there were the comments
to that paper that were published with it and if you read that one or two of them were like you
still don't see how the integration is doing and then my comment to that was then i think i
called that that mini paper response paper like a deeper integration to physics so i know i didn't
succeed in that i apologize but that was the goal
okay awesome dean
would you like to go anywhere ask anything yeah i i don't even know i know i'm just gonna sit back and
then because well well for me
you've taken a whole bunch of stuff around what a form might turn into based on what signals it's
capable of acting on is that is that a a gross besides being a gross oversimplification can we
start there does that make sense yeah so like yes yes that makes sense basically what what i always
like like talking about models like what you put in what you get out right so what we put into the
model is it's an encoding of of signaling responses it's like a classical like if you see signaling a
b but not much of c and d then you think your cell type a whatever that means head type and then you
adjust your own signaling expressions each cell basically has four signaling molecules um that they
can exchange they can express they can sense and based on different combinations of those they think
they're one or the other um and and that is semi-place encoded um so they have basically from that map
that's what the target morphology is is meaning so what we put in basically is is uh is that encoding
on the generative model um general process how they update that the the the tickling diffusion
constants all that stuff that basically how this information then spreads to each other and then what
we get out essentially is is what kind of shapes like you said can can emerge from that um of course
initially right the initial model was basically you don't interested in how many shapes because you're
you're interested can't do the shape that you put into it so the point of this paper was is
keeping that same target the same kind of like i i um each cell has these options that it can be
based on what it's sensing and then it's all secreting but keeping with that can you get different
shapes just by messing up the way that the information flows in not set of the cell and and
it's being processed around the things that's again this that the inspiration from that comes from the
lab i'm working in where we do a lot of this where we don't don't actually mess up the genetic coding
so we don't mess up the genetic model in elective inference terms but we mess up
how how basically the the state space looks like how the information at each point in time is updated
and that's that's the idea of this
what does it look like to engineer or design a different target morphology

how did you fit the bayesian model or fit the flow descriptor whatever it was in its representation towards the represent towards the um anatomy that's described in figure three yes so um the the the the outcomes that we get like the experiment outcomes here we didn't like design the type of fraud like i said we keep those the same we just basically had some intuition that like some of them turn out to be true of how we could change it based on just the asymmetry that we put in an information flow um how would we design this one um mainly so this is an afric level and again this model itself is not that that's already called this before i joined it so that's not i didn't this didn't do as new but how do you approach this so basically idea is um this is the idea of is fairly biological the idea is that you have genetic encodings where certain transcription factors basically right will based on whatever your sense will then basically activate gene expression in the cells that will then allow to differentiate so cells most of morphogenesis essentially starts off from um from something that i mentioned initially you have some asymmetry in in signaling that there's the start that basically even when you already have the the the unfaillized egg in an animal system there's some asymmetry where that's being stored actively by by by the whole organism of the mother essentially and the base from that asymmetry then you have gradient of already of difference and then so basically some cells will say will sense more of signaling type a and less of b and then from that they will make they will be um certain things will be expressed that will then um change its sulfate to a sub sub sulfate and that and then you have more sulfates and the same thing happens there you have now more asymmetry but starting off in the same one so you again will have sense cells that will sense more and less of some signaling signals signaling ligands that will basically then turn out to um be incorporated through transcription factors differential genetic expression um to differentiation so what this is essentially what this is doing right you're you're encoding what what the codes here are essentially those different uh relationships between transcription fact between signaling receptors transcription factors downstream to genetic code where then basically we don't do proveration again but this is the the first step there's basically one snippet of type morphology one aspect where you have at that point in in this hypothetical morphogenesis experiments you have four different signaling type signaling cell four different signaling ligands um that that are being that being expressed that can be expressed by the cells and based on where you are in the cluster you will see more or less of it and based on that information you will then yourself as a cell start to express more or less of because you basically start differentiating um so then you will start expressing more and less of that at some point of that you will reach a certain threshold where they actually will fully have undergone the differentiation process and have a new cell fate and that's what these different um head tail and stuff so of course like you know there's no animal system dino off that has a you know eight cells and uh four cell types it makes a whole body that would be pretty crazy although there are some interesting um there's some interesting model owners and systems where there are um the these um the seal against these worms are one example where they have a a complete map from from from cell to cell from basically a gene they know exactly which cell base it turns out um to be what cell at the end so they can go from the embryo and they can tell exactly this cell is going to be at that at the end of time um this is the

minority of examples most uh sample most model organisms i know are a lot more messier than that and that's what this um trying to basically answer given that you have a lot of mess here that but cells can be different things and then if perturbations happen like they happen all the time in environment and often we do see then you know misformed embryos a lot that we start in the lab other labs to it too we see it in nature all the time where you get completely different um uh outcomes in the morphology even though not all of these occur because of some genetic mutation that's not always the case sometimes it's just basically some perturbation environment and the same genetic code with some little differences that which here is basically the the differences here between one simulation and different if you did it that way we didn't but if you did at least not in this paper if you had different randomization initially of the prior beliefs you can also get different outcomes but um even with one randomization of of the initial beliefs you will get certain outcomes for those cell types based on um on the coding but also on what the environment does that's the kind of the the goal is you have cell types you have you want to see what how can you change the the line of cell type differentiation based on an interplay not just like saying you do this and you tell the cell to do this but based on how they interact with each other hope that answered your question
it makes me think of waddington's landscape the epigenetic landscape which i'll copy in how would you say it relates to epigenetics or the epigenetic
oh sorry if you have my shirt here i have a shirt with it but i don't see it right now yeah i love the uh what incident yeah it's exactly um right so that that was a the big i love you big nap um i'll put it onto a slide yeah basically the right the whole the whole evolution was initially when we started understanding genetic information that it was thought oh well that that's that's the end of it now we know we're done here like we know basically that there's some certain signals that will be assessed and then you will get certain genes but the thing rolls down um but basically just from that it's all all straightforward what was new then later on was the epigenetic part of it there was basically actually modifications happening not on the like you did not not the mutations of the genome but which genes were more or less accessible for expression which you know can so basically you kept the coding the same but based on modifications life system modifications right the the way that the chromosomes where the dna is wrapped around itself in chromosomes the more or less tightly they are called to each other the more or less accessible to transcription factors um and uh that started basically happening later on and now we think a lot more about this and we even see there's even that's even heritable um and uh it's it's it's also interactive with a lot of more the physical forces of it so you know perturbations in the in the mechanical or biological environment will lead to to a lot of fast epigenetic transformations of course it's not a faster it's also important to mention that this a lot faster process it's faster to basically modify some of the proteins that are make up the histones and then make up the accessibility of the genetic code and to start off going some evolution experiment where you start to do it and mutating and cross over until you get the right combination so it's a different layer of information flow that is interactive with genetic level um so what this paper kind of uh interacts with a little bit or you know intersects with it is the

aspect of of health in that initial kind of formation where you have the same genetic code within basically one lifetime how do certain modifications in in sensory precision which you can think of right i didn't make that explicit claim i hate to make claims that i'm not actually then afterwards answering on some relevant biological mechanism but i could think of easily that way that you this these modifications of the the the um the response function um the the sense the sensitivity of the of the double tail experiments and the miss the reduced sensitivity and the cancer initiation those could easily be thought of as as you know short-term epigenetic modifications and we do see that a lot this happens especially in stress environments and embryos and stressful environments in embryonic genesis where where there's very quickly changes that can lead to drastic outcomes and it's it's thought of as a as a quick response mechanism to two basic perturbations and and it's involved as of course in cancer as well that's that's well known so that's kind of how i would think about that just the landscape here is can be thought of as the variation of financial landscape in the sense of like what different outcomes you can get how how how quickly to update your your probability based based on the changes in the environment awesome thank you dean
when i was reading this um i i was thinking of a company that i'm invested in that has taken stem cell derived isolates because people have diabetes and they have created an environment to protect against the perturbation so instead of injecting those pancreatic cells directly into the portal vein they try to uh they have working on trying to sort of create the environment in which the isolates survive so it so essentially it's again rather than looking from the bottom up it's it's sort of creating that safe environment in which these isolates can then carry out their function and so that's what i was kind of thinking about throughout when i was reading your paper now i'm not sure if that again whether i was thinking of the correct analogy but but yeah um in terms of what the pancreatic isolates can do depends on their survivability so if you don't have the right kind of environment for that they tend to die or worse they don't die they become cancerous so it's kind of interesting that you're looking at it here and then i i know of some sort of practical application where the research is really trying to build these environments so that they can actually protect against the perturbations
yeah no that's exactly the long lines that i'm thinking as well
i do but i had like two more minutes um if i get and then i have to push and go
um really appreciated you joining if it works for you to drop in or out at any point in the dot two please feel welcome and any other time you or anyone else are welcome to participate but thanks a lot for joining you're welcome my pleasure see you later okay bye bye thank you excellent that was great fantastic great yeah fun times um okay so here's here's one thought on the relationship with foraging kind of picking up on what we were just looking at so he described some of the manipulations as kind of like interventions that were epigenetic so it wasn't the addition of um like another changing the model structure like changing there from four to five transcription factors um it was just changing their state it's like picking one cell up and moving it would be changing its position but then there's other levers of intervention like changing its sensitivity or other aspects of its cognitive or its generative model and the whole point with science and active inference though see anticipating brain is not a

scientist by brunberg's paper so it's not exactly the same thing but it's about informed or directed experimentation and that's true from the epistemic foraging with the eye circadian to the scientific scientific experiment um when we make targeted interventions into systems based upon a good understanding of them we can learn about the cognitive model that's underlying it or the as if cognitive model so there's a lot of interventions that would not be informative about morphogenesis like jokingly you know hitting the petri dish with a hammer it's it's an intervention that does kill cells it kills cancer but of course isn't that the joke right like there needs to be more to it in the experimental design and in the the measurement of the outcomes so that we're actually learning something and so to make increasingly nuanced designs and learn more and more and apply better and better that's where having like this kind of formal modeling really matters so this is kind of taking just saying like well changes in receptors might be a precursor or a biomarker or an early mechanistic warning of x disease you'll read that a hundred thousand times in the molecular biology literature and this is picking up there and putting it into a modeling architecture where we actually can talk about changes in signaling and position in an integrated way so that it might be able to actually model real settings just like people use the epigenetic landscape heuristically but also increasingly quantitatively to model cell fate decisions but just to kind of close that again it's the interventions that we as the investigator now based upon our understanding of like the natural history of the system that give us updates on our cognitive model of the system so like an ant foraging example is they'll take a desert ant that forages alone the estimate goes out without following like just simply a fair amount trail and pick the ant up when it's like let's just say 30 meters away from the nest or 10 20 meters away and then move it to like 90 degrees rotated the same distance so one extreme case would be it continues from that location a hundred percent as if it were walking back from the north let's just say that would be a pure direction step integrator model there might be an angle it takes between that and for example heading towards the nest that might reflect the usage of other sensory or cognitive features like detecting the polarization of light or scene memory or other ways that it might be able to adjust and turn its vector to some other direction so that's the kind of experiment where we can start to learn like how much are the the nest mates in their foraging adjusting between their own onboard memory versus these external cues that might be picked up like in one instance by any nest mate in that location so there's some cells where moving it into a different tissue the first thing it would do is die because it's just like too acidic or just like not a good environment for it but there are certain interventions that do seem to happen that lead to at the very least we can say states we don't prefer or just more directly like states that are abnormal or unhealthy
um though it really does come back all to blue's question about death and aging and how is that related and what is this living phase between the developmental part and the death part even though there's such a complex continuum between them stephen can i just ask a question on what you're just saying that daniel um do you feel that or do you think that the um the the level of what's happening with epigenetics is fairly flat and it and the once these other types of changes these types of changes happen that the adjacencies permeate out and can the the cells are kind of finding their place um and is that a

bit different to the nestedness you know that we're talking about the nestedness is because um like for instance a tissue or is that also does it then fall into some other type of um behavior where it's like into this idea of the nested um the nested kind of markov blankets and is it then a case of what was the types of information flows which were conservation at one scale have become noise
that makes sense so maybe i'm just curious about if that applies in here
um it very well might i mean what it just makes me think about is the eye specification in root flies and and similar in other insects so this is like an example of a developmental genetics pathway at a relatively downstream level being worked out through a lot of work but these cells are pre-committed to being eye precursor cells but then they undergo a second or a future stage of differentiation that results in a compound eye and then it's the modularity of this developmental framing of first eye precursor cell and then to a specified eye compound cell that lets like the insect eye scale in different ways for example than the mammal eye developmentally so over evolutionary time there's different affordances for insect eye evolution versus mammalian cell eye evolution um because they're on different pathways so it's kind of fractal because this eye field precursor it's not the second one that comes out of the embryo so there's multiple like fractal layers and pre-committals and it's not possible to jump from a canyon way at the bottom directly over there's a few cancers cell papers that do a little bit more of like a reversion and around or a perturbation that pushes and then changes to like susceptibility or kind of like changes in the elevation of these hills between so it's a landscape which isn't just simply flat um it has some scale and then the relationship between how jagged it is and how much noise or flow there is is about how navigable that landscape is in that one model blue
so i have a question for daniel and dean specifically um so when franz was here he talked about the block matrix operator that he used in equation 39 and you guys well maybe it was daniel specifically we're talking about some kind of housekeeping term in the dot zero and i couldn't remember or place the housekeeping term that you were referring to and i wondered if it was like the same or similar in a similar or used in a similar way i think it's connor i'll bring in a housekeeping term slide it was from 32 live stream 32 stochastic chaos markov blankets and um here is the slide in slide 45 here
it was um basically the helmholtz decomposition is usually discussed in terms of just the gradient the solenoidal so sort of like directed term and then the isocontour circulation change in elevation ruler on the hill and then around the hill on the isocontour and then this paper 32
brought in the housekeeping term and then there was like a supplement so we didn't really totally go into it but together um they represent the total flow so it's a slight kind of um fristian variation on the helmholtz decomposition in this context but as connor mentioned in 32.3 it also has like sort of antecedents and maybe even similarities with other um areas so totally remains to be a little bit unpacked but it's the influence of how movement changes the landscape at least that's a little bit how we were framing it like change on a trampoline where the movement is going to influence it you can't just snapshot the landscape and then calculate what would happen and what the gradient and the solenoidal flow would be at every point because it's going to be the future that might be

related to what franz was saying because he said he used it to smooth out the landscape because
otherwise you get stuck in like a little local pocket when you're talking about like you know
something that that's bumpy so it sounds like it could be a similar type of um like housekeeping
thing or just like a smoothing smoothingness when you're when you're going into the bowl you don't
want to get stuck in a bump down along the way right if you're a marble traveling down like if it's
thought it was a cleanse there's just too many too many jokes to be made um but um wow i think in our
last you know 10 ish minutes we should just chill land the plane prepare for dot two it was pretty
unexpected and awesome to have the um time that we got to speak and i think that's like the housekeeping
term in action which is a process and a protocol and then people who have different perspectives on
a situation but the group can always readapt to a changing landscape and so even if that's interpreted
as structure learning at a higher level from the sort of agent view looking up it's equivalent to
adapting to a different situation in some fractal way slightly different or very different using the
same affordances and perceptive and cognitive features as it had in the time step before and
that's why um this model that looks at the flow of the autonomous states so um the states that can
actually be controlled in the markov blanket partition
in 30 by defining a specific function that we want to like be fitting well by really focusing on the flow
over the blanket and internal states it allows inference on that component perception cognition and action
of the given modeled entity to be uncoupled in a certain way from inference on the hidden external state evolution
which is fundamentally unknowable but in a slightly different way
if that makes any sense
and a classic moment blue to suggest the the mitotic elements and like
just to bring in the continuity and then the way that that gives rise to symmetry breaking sort of for free
um and that this and
so it's
cool conversation that we had today and so
so i did up as a dot one might
i didn't get to mention and i was going to mention it um but like you know there's that fractal
fmri brain pattern paper that i'm like dying to read and um would love to discuss with the authors also
um but it's interesting just in terms of the the fractal dimension also that franz was talking about
it's super cool
um
is there anything else that we want to like um
write down to think about in the dot two or questions that we'd like to ask franz if he returns or
just things that we want to
take moving forward in the dot two already
um
i'm still that one

yeah i i agree it's like it's like you're eating dinner it's like what do you want dinner tomorrow
inverting space and time we didn't really get to um and some of these initial points it was
right least action math and morphology and then
generalized flow then taking us all the way back dean so i i would like to maybe ask ask franz about
quantum mechanics and what role if any that that played in um like what is there a quantum perspective
that that he was perhaps viewing in from um when he was writing this paper i would like to ask just
that one question because that will encompass inverting space and time and it will encompass
um what we're talking about with the probability current and um the the probability mass conservation
so like that all is like bridging into the quantum where we'll go in the 40. so i think i would like to ask
that one thing dean what are you like looking forward to or would you want to add here at the end
i'm just going to go back to the drawing board and basically look look a little more closely at some of those
math operations because i don't tend to give those as much attention as i need to i think it's really interesting
now that we've had a chance to sort of bring more people into the conversation i think
think it uh there's three parts to that i think we can agree there are three
things that we can really center on now there's the the formally form forming which is what figure
four and five do but as soon as you go to figure five what's interesting is is that the
the attention seems to be on the shape that the cells eventually take and yet in the in the sort of
in the blackness around each one of those moment captures there's there's signaling and there's changing
right there's so there's all the part we've talked about today which is the whole flowing piece and
what where is it going and how is it channeling and how is it narrowing or how is it spreading and
how is it dissipating and all that stuff but in those even in those diagrams it's hard to figure out
where the signal is even when they introduce arrows they introduce the arrow pointing at the form
not the necessarily the signaling and the changing and that's where the math really comes in i think
because without the math you you lose sight of those other two present factors that are you know all
moving and mixing and and turning into something so i i hope france comes back because man he's got a
heck of a lot more um he's got a lot more to tell us about really because this is such a dense paper like i
i think i mentioned to you daniel before before the point zero this could be a 13-week course this one
paper if you really wanted to i don't know how many hours he spent on this but he already admitted that
going to carl's lab and and and from the lab he's at there's way more multiples of 10 000 hours
of work that were done before he started typing things out so yeah
awesome with that made me like think about is this idea of like multiple kinds of invisibility or overlay
or unpacking like projection up into a bigger dimension of interpretation like you pointed out
how the like the focus was on morphology which is to say the final realize like this picture is of the
morphology of the animal it's not of the signaling density but this could be like presented as just a
gradient of transcription factor a or it could be presented as a gradient of vitamin a or like it's
kind of like carillion photography like looking at the morphology with the energy field i don't know
if that's exactly what it is but that sort of idea with the field based perspective but then it gets
hard to show many overlaying fields because like what do you do 20 overlaying colors but we we can't

actually see that and so that's just kind of interesting like a question about visualization of
higher dimensional models with a lot of overlays and then that's like one kind of invisibility which
is something that's modeled but just not graphically shown easily like the density of 50 pheromones or 50
transcription factors or vitamins but then also but but it does exist and it's like modeled as an actual
chemical component of the biological system like the generative process and then there's this generative
model with the mathematical derivatives the generalized coordinates of motion of cellular position and
those are also like invisible in a different way because they're um a modeling tool and the derivatives are
not anywhere in reality like what is where's the seventh derivative of the baseball's movements
just relative to what where is it hiding in the current moment this is just a purely tool driven
way of thinking about the current moment not just in terms of its composition and like anatomy and position
and the lowest level of the observed model but like these higher levels which are real yet not existing
in the moment structurally real in the structural realism sense i'll take your word for it i've been
wanting to say that for a long time i have to now go back and figure it out so um but
what a discussion i hope like for those who stick around to the end here that they enjoyed this that this
wasn't just um total uh total morpho pastrophe
all right well fun times and awesome to have all the good discussions and appreciate everybody who
was here blue and steven and dean so see you all next week or any other time peace thank you bye thank you
bye
you